

SAI KRISHNA YEMINENI

AI/ML ENGINEER

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SUMMARY

AI/ML Engineer with 4+ years of experience developing production-grade AI solutions across healthcare and consulting environments. Experienced in **Agentic AI**, **Generative AI**, **LLM** applications, **RAG** pipelines, **NLP**, predictive modeling, and scalable machine learning systems. Skilled in building AI products using **Python**, **PyTorch**, **TensorFlow**, **LangChain**, **Hugging Face Transformers**, **AWS (SageMaker, Lambda)**, **Docker**, **Kubernetes**, and **Apache Spark**. Proven ability to translate business and clinical requirements into secure, reliable, and measurable AI solutions.

TECHNICAL SKILLS

Languages:	Python, R, C++, SQL, Bash scripting
Generative AI & LLMs:	Agentic AI, Generative AI, Large Language Models, RAG, Prompt Engineering, LangChain, Hugging Face Transformers, Vector Databases, Embeddings, AI Agents, LLM Evaluation
ML Frameworks:	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, SciPy, ggplot2, Seaborn
Data Visualization Tools:	Tableau, Power BI, Advanced Excel, Statistics,
Functional Expertise:	Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transformers, GANs
Database:	MySQL, MongoDB, PostgreSQL, SQL Server
Model Development:	Supervised/Unsupervised learning, Time series forecasting, Anomaly detection, Reinforcement Learning
Cloud Platform:	AWS (SageMaker, Lambda, EC2), GCP, Microsoft Azure (ML Studio, Azure Databricks)
Optimization Techniques:	Hyperparameter tuning, Grid search, Randomized search, Bayesian Optimization, Genetic Algorithms
Deployment & CI/CD:	Docker, Kubernetes, MLflow, TensorFlow Serving, Flask/Django for model APIs
AI Product Development:	AI product lifecycle, use-case discovery, rapid prototyping, model evaluation, API integration, user feedback analysis, A/B testing, monitoring, and responsible AI
Big Data Tools:	Apache Spark, Hadoop, Dask, Kafka, Hive

PROFESSIONAL EXPERIENCE

Johnson & Johnson, MD, USA | AI/ML Engineer May 2024 – Current

- Designed and deployed an **Agentic AI**-based clinical risk detection workflow using **Python**, **AWS SageMaker**, and event-driven services, enabling automated data validation, risk scoring, and alert generation for incoming **EHR** records and reducing alert latency by **27%**.
- Built patient risk stratification models using **TensorFlow** and **XGBoost**, improving prediction accuracy by **19%** through feature engineering, cross-validation, and continuous retraining.
- Developed a **RAG**-based clinical document assistant using **Python**, **Transformer models**, **Embeddings**, and **Semantic Retrieval** to generate structured summaries, reducing manual document review from **10 hours to 40 minutes**.
- Built a multi-step **Agentic AI** workflow for document classification, information extraction, retrieval, response generation, and validation, with human review checkpoints for low-confidence outputs.
- Created an **LLM** evaluation framework to measure retrieval relevance, response completeness, hallucination risk, latency, and output consistency using curated clinical test cases and human feedback.
- Partnered with clinical, engineering, and product stakeholders to define **AI** use cases, success metrics, user workflows, and model acceptance criteria, converting prototypes into production-ready **AI product** features.
- Deployed containerized **LLM** and **NLP** services using **FastAPI**, **Docker**, **AWS Lambda**, and **GitHub Actions**, enabling scalable integration with clinical applications.
- Processed more than **3 TB** of healthcare data daily using **Apache Spark**, **Apache Hadoop**, and **Apache Hive** to support analytics, model training, and production inference.

Deloitte, India | AI/ML Engineer Dec 2020 – July 2023

- Built an intelligent clinical anomaly detection engine leveraging **PyTorch** and neural networks, reducing false positives by 31 units and enhancing early risk sensitivity across 22 hospital care ecosystems.
- Enhanced rare-event clinical classification accuracy using **XGBoost** and **SMOTE**, improving predictive reliability for 18 high-risk patient patterns across population health operations and digital care platforms.
- Automated ingestion of 80 million daily patient records via **Apache Spark**, **Kafka**, and **Airflow**, enabling scalable ETL pipelines for feature enrichment and retraining of over 20 clinical models.
- Developed NLP entity extraction and intent classification modules using **BERT** and tokenizer tuning, automating parsing of 1.2 million EHR documents per quarter with minimal manual review.
- Containerized 10 real-time inference APIs using **Docker**, reducing deployment cycle time by 60 units and maintaining 99.99 uptime through **AWS SageMaker** and integrated **ECR** workflows.
- Implemented automated CI/CD pipelines using **MLflow** and **FastAPI**, mitigating 25 model drift incidents and ensuring reproducibility across 6 environments with centralized clinical experiment metadata.
- Visualized model-driven clinical risk indicators using **Plotly** and **SHAP**, accelerating 45 compliance investigations and cutting auditor validation effort by 200 analyst-hours per reporting cycle.
- Designed **Tableau**-based executive dashboards integrating AI-powered care risk summaries, delivering real-time insights on 9 regions and improving visibility into 600 thousand flagged patient cases quarterly.

EDUCATION

Master of Science in Computer Science - Rivier University, Nashua, NH, USA	Sept 2023 – May 2025
Bachelor of Technology in Computer Science & Engineering - KL University Andhra Pradesh, India	Jun 2016 – Sept 2020

CERTIFICATION

- AWS Certified AI Practitioner** – Amazon Web Services (2026–2029)
- AWS Certified Machine Learning – Specialty** – Amazon Web Services (2026–2029)
- Google Cloud Certified – Generative AI Leader** – Google Cloud (2026–2029)